

# Quantum Fault Tolerance Meets Toom's Contours

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## 1 Hadamard and S Gates

**Claim 1.1.** For any generator subset  $S'$  of size  $d'$ , denote by  $x_{S'}$  and  $z_{S'}$  the following assignments:

$$x_{S'} := \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S'} \quad z_{S'} := \sum_{u \in \langle S/S' \rangle} \theta_{\mathbf{u}, S'}$$

Then  $x_{S'}$  and  $z_{S'}$  are binary representations of anticommuting logical codewords of the  $(d, d')$ -Toric code. In particular,  $x_{S'} \in \mathcal{C}_X / \mathcal{C}_Z^\perp$ ,  $z_{S'} \in \mathcal{C}_Z / \mathcal{C}_X^\perp$ , and  $x_{S'} z_{S'} = 1$ . In addition, let  $X_{S'}$  and  $Z_{S'}$  be the Pauli matrices obtained by multiplying each non-trivial coordinate of them by  $X$  or  $Z$ , respectively.

*Proof.* By definition  $x_{S'}$  and  $z_{S'}$  are binary assignments on the  $d'$ -dimensional faces. We start by showing that  $x_{S'} \in \mathcal{C}_X$  and  $z_{S'} \in \mathcal{C}_Z$ :

$$\begin{aligned} \partial^- \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S'} &= \sum_{u \in \langle S' \rangle} \sum_{g \in S'} \theta_{\mathbf{u}, S'/g} + \theta_{\mathbf{g}\mathbf{u}, S'/g} \\ &= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S'/g} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{g}\mathbf{u}, S'/g} \\ &= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S'/g} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S'/g} = 0 \\ \partial^+ \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S/S'} &= \sum_{u \in \langle S' \rangle} \sum_{g \in S'} \theta_{\mathbf{u}, S/S' \cup g} + \theta_{\mathbf{g}^{-1}\mathbf{u}, S/S' \cup g} \\ &= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S/S' \cup g} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{g}^{-1}\mathbf{u}, S/S' \cup g} \\ &= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S/S' \cup g} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S/S' \cup g} = 0 \end{aligned}$$

Thus  $x_{S'} \in \mathcal{C}_X$  and  $z_{S'} \in \mathcal{C}_Z$ , now for showing that they are not binary representations of stabilizers, it's enough to show they admit no vanishing product.

$$x_{S'} z_{S'} = \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S'} \sum_{u \in \langle S/S' \rangle} \theta_{\mathbf{u}, S'} = \theta_{\mathbf{e}, S'} \theta_{\mathbf{e}, S'} = 1$$

□

**Definition 1.2** (The Transpose Map). Let  $d, d'$  be integers, and let  $\mathcal{G}$  be a  $(d, d')$ -Toric complex generated by the generator set  $S$ . We define the *transpose map*  $\phi : \mathcal{F}^{d'} \rightarrow \mathcal{F}^{d-d'}$  by acting on the basis elements as follows:

$$\phi(\theta_{\mathbf{v}, \mathbf{S}'}) \rightarrow \theta_{\mathbf{v}^{-1}, \mathbf{S}/\mathbf{S}'}$$

Notice that for  $d' = d - d'$ , the transpose map is a permutation over the bits.

**Definition 1.3** (Hadamard-Type Gate  $H_\phi$ ). For  $d' = d - d'$ , we define the logical Hadamard-type gate  $H_\phi : L \rightarrow L$  as follows, for any  $S' \subset S$ , of size  $d'$ :

$$\begin{aligned} H_\phi(X_{S'}) &\rightarrow Z_{S/S'} \\ H_\phi(Z_{S'}) &\rightarrow X_{S/S'} \end{aligned}$$

**Claim 1.4.** The Hadamard-type gate  $H_\phi$  has a fold-transversal implementation by, first applying  $H^n$  over all the physical qubits, and then permuting them by the transpose map  $\phi$ , namely  $\hat{H}_\phi = \phi \circ H^n$ .

*Proof.* Clearly, for any subset of generators  $S' \subset S$ , the map  $\phi$  sends  $x_{S'}$  to  $z_{S/S'}$ :

$$\phi(x_{S'}) = \phi\left(\sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, \mathbf{S}'}\right) = \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, \mathbf{S}/\mathbf{S}'} = z_{S/S'}$$

Moreover,  $\phi$  sends binary representations of  $Z$ -stabilizers to binary representations of  $X$ -stabilizers, and vice versa. To see this, consider a  $d' + 1$  dimensional face, and consider the image of the stabilizer it defines.

$$\begin{aligned} \phi(\partial^- \theta_{\mathbf{v}, \mathbf{S}'}) &= \phi\left(\sum_{g \in S'} \theta_{\mathbf{v}, \mathbf{S}'/\mathbf{g}} + \theta_{\mathbf{g}\mathbf{v}, \mathbf{S}'/\mathbf{g}}\right) \\ &= \sum_{g \in S'} \theta_{\mathbf{v}^{-1}, (\mathbf{S}/\mathbf{S}') \cup \mathbf{g}} + \theta_{\mathbf{g}^{-1}\mathbf{v}^{-1}, (\mathbf{S}/\mathbf{S}') \cup \mathbf{g}} \\ &= \partial^+ \theta_{\mathbf{v}^{-1}, \mathbf{S}/\mathbf{S}'} \end{aligned}$$

And since  $\phi \circ \phi = I$ , we also have that  $\phi(\partial^+ \theta_{\mathbf{v}^{-1}, \mathbf{S}/\mathbf{S}'})$ . Thus,  $H^{\otimes n} \circ \phi$  sends  $Z$ -stabilizers to  $X$ -stabilizers and vice versa, thereby preserving the code space. Combining this with the above,  $H^{\otimes n} \circ \phi$  realizes the logical form  $H_\phi$  in the  $(d, d/2)$ -Toric code.  $\square$

**Corollary 1.5.** For the  $(d, d/2)$ -Toric code, consider the gate  $CZ_{S' \rightarrow S/S'} : L^{\otimes 2} \rightarrow L^{\otimes 2}$ , defined to act on two encoded registers by applying  $Z_{S/S'}$  on the second register, controlled by whether the logical  $X_{S'}$  is turned on in the first register. Then  $CZ_{S' \rightarrow S/S'}$  has a fold-transversal realization.

*Proof.* Consider the following circuit,

1. Apply  $H^n$  on the second register.
2. For any bit  $\theta_{\mathbf{v}, \mathbf{S}'}$ , apply  $CX$  from  $\theta_{\mathbf{v}, \mathbf{S}'}$  in the first register into  $\phi(\theta_{\mathbf{v}, \mathbf{S}'})$  in the second register.

3. Again apply  $H^n$  on the second register.

□

**Definition 1.6** (The  $\varphi$  Rotation). Consider a permutation from the generator set to itself  $\pi: S \rightarrow S$ . The permutation  $\pi$  induces a natural isomorphism by substituting generators of the product representation with their corresponding images according to  $\pi$ . We define the  $\varphi_\pi$  rotation,  $\varphi_\pi: \mathcal{F}^{d'} \rightarrow \mathcal{F}^{d'}$ , to act on the  $d'$ -dimensional faces as follows:

$$\varphi_\pi(\theta_{\mathbf{v}, \mathbf{S}'}) := \theta_{\pi(\mathbf{v}), \pi(\mathbf{S}')}$$

**Claim 1.7.** For  $d' = d - d'$ , fix a generator subset  $S'$  of size  $d'$ , and consider a permutation  $\pi: S \rightarrow S$  such that  $\pi(S') = S \setminus S'$ . Then  $\varphi_\pi$  satisfies the following:

1.  $\varphi_\pi$  sends the binary representation of  $X$ -stabilizers to themselves, and similarly for the binary representations of the  $Z$ -stabilizers.
2.  $\varphi_\pi$  sends  $x_{S'}$  to  $X_{S/S'}$  and vice versa.

*Proof.* First, notice that we require the condition  $d' = d - d'$  to define  $\pi$ ; otherwise, no permutation  $\pi$  satisfies that  $\pi(S') = S/S'$ . For the first part of the claim, consider an arbitrary subset of generators  $J \subset S$  of size  $d' + 1$ , and notice that for any generator  $g \in J$ , we have  $\varphi_\pi(J/g) = \varphi_\pi(J)/\varphi_\pi(g)$ . Thus:

$$\begin{aligned} \varphi(\partial^- \theta_{\mathbf{v}, \mathbf{J}}) &= \varphi\left(\sum_{g \in J} \theta_{\mathbf{v}, \mathbf{J}/g} + \theta_{\mathbf{g}\mathbf{v}, \mathbf{J}/g}\right) \\ &= \sum_{g \in J} \theta_{\varphi(\mathbf{v}), \varphi(\mathbf{J}/g)} + \theta_{\varphi(\mathbf{g})\varphi(\mathbf{v}), \varphi(\mathbf{J}/g)} \\ &= \sum_{g \in \varphi(J)} \theta_{\varphi(\mathbf{v}), \varphi(\mathbf{J})/g} + \theta_{\mathbf{g}\varphi(\mathbf{v}), \varphi(\mathbf{J})/g} \\ &= \partial^- \theta_{\varphi(\mathbf{v}), \varphi(\mathbf{J})} \end{aligned}$$

Similarly, we find that for any subset of generators  $J \subset S$  of size  $d' - 1$ :

$$\begin{aligned} \varphi(\partial^+ \theta_{\mathbf{v}, \mathbf{J}}) &= \varphi\left(\sum_{g \in S/J} \theta_{\mathbf{v}, \mathbf{J} \cup g} + \theta_{\mathbf{g}^{-1}\mathbf{v}, \mathbf{J} \cup g}\right) \\ &= \sum_{g \in J} \theta_{\varphi(\mathbf{v}), \varphi(\mathbf{J} \cup g)} + \theta_{\varphi(\mathbf{g}^{-1})\varphi(\mathbf{v}), \varphi(\mathbf{J} \cup g)} \\ &= \sum_{g \in S/\varphi(J)} \theta_{\varphi(\mathbf{v}), \varphi(\mathbf{J}) \cup g} + \theta_{\mathbf{g}^{-1}\varphi(\mathbf{v}), \varphi(\mathbf{J}) \cup g} \\ &= \partial^+ \theta_{\varphi(\mathbf{v}), \varphi(\mathbf{J})} \end{aligned}$$

So, we got that  $\varphi_\pi$  sends binary representations to representations of the same type.

$$\varphi_\pi(x_S) = \varphi_\pi\left(\sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, \mathbf{S}'}\right) = \sum_{u \in \langle S/S' \rangle} \theta_{\mathbf{u}, \mathbf{S}/\mathbf{S}'} = x_{S/S'}$$

□

**Definition 1.8** (The Nice Encoding). Consider  $(d, d/2)$ -Toric code and fix a subset  $S' \subset S$ , at size  $d'$ . We define the *nice encoding* to be the subsystem code, which encodes a single logical qubit as follow:

$$\begin{aligned} |\bar{0}\rangle &= |\mathcal{C}_Z^\perp\rangle \\ |\bar{1}\rangle &= |x_{S'} + x_{S/S'} + \mathcal{C}_Z^\perp\rangle \end{aligned}$$

**Claim 1.9.** Given a resource state, the nice encoding has a fold-transversal realization of the logical **S** gate.

*Proof.* Consider the resource state defined by:

$$|\mathbf{S}\rangle = \frac{1}{\sqrt{2}} (|\bar{0}\rangle + i |\bar{0} + x_{S'}\rangle)$$

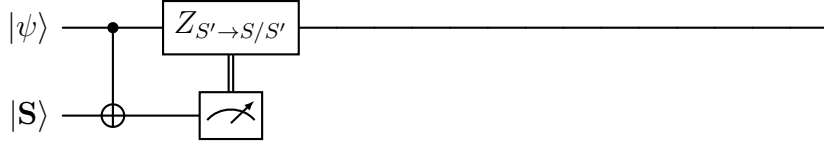


Figure 1: Simulating the **S**-gate, using the  $|\mathbf{S}\rangle$  magic state.

$$\begin{aligned} |\psi\rangle |\mathbf{S}\rangle &= (\alpha |\bar{0}\rangle + \beta |\bar{0} + x_{S'} + x_{S/S'}\rangle) \frac{1}{\sqrt{2}} (|\bar{0}\rangle + i |\bar{0} + x_{S'}\rangle) \\ &\mapsto \frac{1}{\sqrt{2}} (\alpha |\bar{0}\rangle |\bar{0}\rangle + i\alpha |\bar{0}\rangle |\bar{0} + x_{S'}\rangle \\ &\quad + \beta |\bar{0} + x_{S'} + x_{S/S'}\rangle |\bar{0} + x_{S'} + x_{S/S'}\rangle + i\beta |\bar{0} + x_{S'} + x_{S/S'}\rangle |\bar{0} + x_{S/S'}\rangle) \end{aligned}$$

□

## 2 Toric Encoded Circuits

**Definition 2.1** (Toric Encoded Circuit). Let  $C$  be a quantum circuit. We say that  $C$  is *Toric encoded* if the following holds:

1. There is a Toric code  $\mathcal{C}$  such that the qubits of  $\mathcal{C}$  can be partitioned into registers  $R_1, \dots, R_k$ , and each of the registers is encoded by  $\mathcal{C}$ .
2. The gates in  $C$  with even time coordinates are sweep decoding gates.

For a set of faults  $\mathcal{E}$  the gates in  $C[\mathcal{E}]$  can be decomposed to cycles, of the following form:

$$C = D_l P_l \Lambda_l, \dots, D_1 P_1 \Lambda_1$$

**Definition 2.2** (Base graph  $G_o$ ). Let  $C$  be a quantum circuit, at depth  $d$ , with registers  $R_1, \dots, R_k$ , all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by  $\mathcal{G}_1, \dots, \mathcal{G}_k$  the Toric complexes that induce the codes. We define the base graph of  $C$ , denoted by  $G_o = (V_o, E_o)$  as follows:

1. The vertices  $V_o$  are tuples, where the first coordinates correspond to a vertex in the union over the vertices in  $\mathcal{G}_1, \dots, \mathcal{G}_k$ , and the second coordinate is associated with a time. Namely:

$$V_o := \bigsqcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges  $E_o$  are derived from the original edges in  $\mathcal{G}_1, \dots, \mathcal{G}_k$ , with the addition that any two vertices in preceding times  $t$  and  $t+1$  that support qubits (faces) such that  $C$  acts non-trivially on those qubits at time  $t$  are also connected by an edge. Namely:

$$E_o := \left\{ \{(v, t), (u, t')\} : \begin{array}{l} \text{either there exists } i \text{ such } \{v, u\} \in \mathcal{G}_i \text{ and } t' = t, t+1 \\ \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1} + 1 \end{array} \right\}$$

**[COMMENT]** Here we should add that a vertex  $v$  belongs to location  $l$  if there is a qubit  $f$  that  $v$  belongs to its associated face.

**Definition 2.3** (Time Graph of  $C(\mathcal{E})$ ). For a quantum circuit  $C$  and set of faults  $\mathcal{E}$ , let us denote the base graph of  $C[\mathcal{E}]$  by  $G_o$ . For each time  $t$ , denote the deviation induced by  $\mathcal{E}$  at time  $t$  by  $\mathcal{D}_t$ , so  $\partial\mathcal{D}_t$  is the syndrome that the qubits in the deviation admit. We define the history graph  $\mathcal{H} = (V_{\mathcal{H}}, E_{\mathcal{H}})$  as follows:

1. The vertices  $V_{\mathcal{H}}$  are taken to be the vertices of  $G_o$ , namely tuples  $(v, t) \in V_o$ , that satisfy that  $v$  belongs to a face associated with a check, which, at time  $t$ , belongs to the  $\partial\mathcal{D}_t$ . Namely:

$$\{v_t : v_t = (v, t) \in V_o \text{ and there exists a check } f \in C \text{ such } v \in f \in \partial\mathcal{D}_t\}$$

2. The edges  $E_{\mathcal{H}}$  is partitioned into two types  $A_{\mathcal{H}}$  and  $F_{\mathcal{H}}$ , where  $A_{\mathcal{H}}$  are the edges in  $E_o$  restricted to  $V_{\mathcal{H}}$  and  $F_{\mathcal{H}}$  are defined to be the . Namely:

$$\begin{aligned} A_{\mathcal{H}} &= \{e = \{u, v\} \in E_o : u, v \in V_{\mathcal{H}} \text{ and } T(v) = T(u) \pm 1\} \\ F_{\mathcal{H}} &= \{e = \{u, v\} \in E_o : \text{exists } w \in V_{\mathcal{H}} \text{ such } \{v, w\}, \{u, w\} \in A_{\mathcal{H}}\} \end{aligned}$$

By definition of the construction,  $G_{\mathcal{H}}$  is a time-graph.

**Definition 2.4** (**Size()**-Function). Let  $S$  be a subset of vertices, we define the *size* of  $S$  to be:

$$\mathbf{Size}(S) := \sum_i^{d+1} \max_{v \in S} L_i(v)$$

**Claim 2.5.** Let  $\mathbf{S} = \{S_1, S_2, \dots, S_k\}$  where  $S_i \subset \mathbb{R}^d$ , introduce an edge between  $S_i$  and  $S_j$  iff  $S_i \cap S_j \neq \emptyset$ . Assume that  $\mathbf{S}$  is connected. Then  $\mathbf{Size}(\cup_{i=1}^k S_i) \leq \sum_{i=1}^k \mathbf{Size}(S_i)$ .

*Proof.* It suffices to prove that if  $R_1 \cap R_2 \neq \emptyset$ , then  $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$ . Let  $a_{i,j} = \max_{v \in R_j} L_i(v)$ . Then for any  $w \in R_1 \cap R_2$  we have:

$$\begin{aligned} \mathbf{Size}(R_1 \cup R_2) &= \sum_i^{d+1} \max a_{i,1}, a_{i,2} = \sum_i^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\ &\leq \sum_i^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\ &= \mathbf{Size}(R_1) + \mathbf{Size}(R_2) \end{aligned}$$

□

**Definition 2.6** (Size Invariant Action). [\[COMMENT\] Here we classify both transversal gates, and rotation on the Toric.](#) An action  $g$  is said size invariant action if, for any assignment  $z$ , it holds that  $\mathbf{Size}(gz) = \mathbf{Size}(z)$ .

**Claim 2.7.** The sweep rule is Size  $\xi$ -shrinking.

*Proof.*

□

[\[COMMENT\]](#) Eventually, we might not use the projected graph, and then we will split to history of the slices lemma for 3 parts. The first is when time =  $3t$ , in that we apply an invariant action so.